1 Solution — GIAM 6.5 Problem 1

1.1 Problem

For each of the following functions, give its domain, range, and a possible codomain.

- **a)** $f(x) = \sin(x)$
- **b)** $g(x) = e^x$
- **c)** $h(x) = x^2$
- **d)** $m(x) = \dfrac{x^2+1}{x^2-1}$
- **e)** $n(x) = \lfloor x \rfloor$
- **f)** $p(x) = \langle \cos(x), \sin(x) \rangle$

1.2 Setup

Recall:

- **Domain.** The set of inputs on which the function is defined.
- **Range** (or **image**). The set of outputs the function actually attains: $\{f(x) : x \in \text{domain}\}$.
- **Codomain.** A target set declared with the function. The codomain must contain the range, but is otherwise a choice — any superset of the range is a valid codomain.

So a "possible codomain" is any set containing the range. Two natural choices stand out for each function below: the range itself (which makes the function surjective by construction), or the broader ambient set ($\mathbb{R}$, or $\mathbb{R}^2$ for vector-valued p).

1.3 a) $f(x) = \sin(x)$

- **Domain:** $\mathbb{R}$.
- **Range:** $[-1, 1]$. The sine function oscillates between -1 (attained at $x = -\pi/2 + 2\pi k$) and 1 (attained at $x = \pi/2 + 2\pi k$), and takes every value in between by

continuity.

- **Codomain:** $\mathbb{R}$ (or $[-1, 1]$).

1.4 b) $g(x) = e^x$

- **Domain:** $\mathbb{R}$.
- **Range:** $(0,\ \infty)$. We have $e^x > 0$ for every real x, with $e^x \to 0$ as $x \to -\infty$ (never reaching 0) and $e^x \to \infty$ as $x \to \infty$.
- **Codomain:** $\mathbb{R}$ (or $(0, \infty)$).

1.5 c) $h(x) = x^2$

- **Domain:** $\mathbb{R}$.
- **Range:** $[0,\ \infty)$. The square of a real number is non-negative, and every non-negative real y is realized as x^2 via $x = \sqrt{y}$.
- **Codomain:** $\mathbb{R}$ (or $[0, \infty)$).

1.6 d) $m(x) = \dfrac{x^2+1}{x^2-1}$

- **Domain:** $\mathbb{R} \setminus \{-1,\ 1\}$. The denominator vanishes exactly at $x = \pm 1$, so those points are excluded.

- **Range:** $(-\infty,\ -1] \cup (1,\ \infty)$.

  Substitute $t = x^2$. As x ranges over $\mathbb{R} \setminus \{\pm 1\}$, t ranges over $[0,\ 1) \cup (1,\ \infty)$, and

$$m \ = \ \frac{t+1}{t-1}.$$

  The derivative

$$\frac{dm}{dt} \ = \ \frac{(t-1)-(t+1)}{(t-1)^2} \ = \ \frac{-2}{(t-1)^2} \ < \ 0$$

  shows m is strictly decreasing on each piece.

- ◦ On $t \in [0, 1)$: $m(0) = -1$, and $m \to -\infty$ as $t \to 1^-$. So this piece sweeps out $(-\infty, -1]$.
  - ◦ On $t \in (1, \infty)$: $m \to +\infty$ as $t \to 1^+$, and $m \to 1$ from above as $t \to \infty$ (horizontal asymptote, never attained). So this piece sweeps out $(1, \infty)$.

  Together, the range is $(-\infty, -1] \cup (1, \infty)$.
- **Codomain:** $\mathbb{R}$.

1.7 e) $n(x) = \lfloor x \rfloor$

- **Domain:** $\mathbb{R}$.
- **Range:** $\mathbb{Z}$. The floor of any real is an integer, and every integer k is its own floor ($\lfloor k \rfloor = k$), so the range is exactly $\mathbb{Z}$.
- **Codomain:** $\mathbb{Z}$ (or $\mathbb{R}$).

1.8 f) $p(x) = \langle \cos(x), \sin(x) \rangle$

- **Domain:** $\mathbb{R}$.
- **Range:** the **unit circle** $S^1 = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 = 1\}$. Since $\cos^2(x) + \sin^2(x) = 1$ for every real x, every output lies on S^1; conversely, every point of S^1 is parametrized as $(\cos x, \sin x)$ for some $x \in [0, 2\pi)$.
- **Codomain:** $\mathbb{R}^2$ (or S^1).

1.9 Remark — Codomain Is a Choice

For each function above, two natural codomain choices stand out:

1. **The range itself.** With this choice, the function is automatically *surjective*.
2. **The ambient space** ($\mathbb{R}$, or $\mathbb{R}^2$ for p). This is the broadest natural target, and the function is then surjective only if its range happens to fill the ambient set — which it usually does not.

Both are valid: codomain is a declared part of a function's definition, subject only to the constraint that it must contain the range. The choice matters precisely when one cares about *surjectivity*, since surjectivity of $f : X \to Y$ is the statement "$\operatorname{range}(f) = Y$."